The Navier-Stokes Bridge: From Geometric Foliation through Holographic Gravity to Biophysical Coherence

Dávid Navrátil
Independent Researcher, Theoretical Biophysics
david.navratil2016@gmail.com

March 2026

Abstract

This paper establishes the Navier-Stokes equations as the fundamental bridge connecting three domains: a first-principles geometric framework, holographic gravity, and the biophysics of microtubules. The framework originates from a single space-filling constraint on a three-dimensional manifold, which uniquely determines a characteristic scaling constant $\eta \approx 1.839$ (the Tribonacci constant) as the dominant eigenvalue of the system's transfer matrix. The maximal Lyapunov exponent, $\Lambda = \ln(\eta) \approx 0.609$, is shown to be numerically identical (within 1.6%) to the sub-Ohmic bath exponent $s \approx 0.6$ independently measured in the hydration shells of microtubules. This identification is the central thesis. The Navier-Stokes equations, which emerge from Einstein's equations on the holographic boundary (Bhattacharyya et al., 2008) and simultaneously describe the viscous dynamics of the hydration shell, provide the unifying language. A discrete Green-Kubo formulation yields an exact viscosity-to-entropy ratio $\eta_{\text{visc}}/s = \eta/(\eta - 1) \approx 2.192$. The model makes testable predictions for gravitational wave ringdown, fast scrambling, and a running spectral dimension consistent with Causal Dynamical Triangulations. A k-nacci generalization predicts a universal upper bound $\ln(2) \approx 0.693$ for the viscosity parameter.

1 Introduction

1.1 The Three Domains

The holographic principle [1], the fluid-gravity correspondence [2], and the tensor network interpretation of spacetime [3] all point to a deep connection between the structure of space and the dynamics of matter. The idea that black hole horizons behave as viscous fluids, first proposed in the membrane paradigm by Damour [4] and later made precise by the fluid-gravity duality, finds a concrete realization in this work.

The central thesis of this paper is that the Navier-Stokes equations form a fundamental bridge connecting three seemingly disparate domains:

1. A first-principles geometric framework of dimensional foliation, governed by the Tribonacci constant $\eta \approx 1.839$.

2. Holographic gravity, where the boundary dynamics of an AdS-Schwarzschild black hole are described by viscous hydrodynamics.

3. The biophysics of microtubules, where the hydration shell acts as a sub-Ohmic thermal bath with anomalous transport exponent $s \approx 0.6$ [5].

1.2 The Central Identification

The key to this bridge is the identification of a single dimensionless parameter that governs the dynamics in all three domains: the anomalous transport exponent. In the geometric framework, this is the maximal Lyapunov exponent $\Lambda = \ln(\eta) \approx 0.609$. In the biophysical context, this is the sub-Ohmic bath exponent $s \approx 0.6$, which governs the anomalous diffusion $\langle r^2(t) \rangle \propto t^s$ and the viscoelastic response of confined water. The remarkable numerical agreement between these two independently derived parameters, $|s - \ln(\eta)|/\ln(\eta) < 1.6\%$, is the empirical foundation of this work.

This paper is structured as follows. Section 2 introduces the geometric foundation. Section 3 explores symmetry, conservation, and information flow. Section 4 details the holographic realization and thermodynamics. Section 5 establishes the Navier-Stokes bridge as the central result, including a detailed table of numerical concordances. Section 6 presents testable predictions. Section 7 generalizes the framework to a k-nacci hierarchy. Section 8 discusses the dilaton coupling. Section 9 concludes.

2 The Geometric Foundation

2.1 The Space-Filling Constraint and the Tribonacci Constant

Principle 1 (Space-Filling Constraint). A three-dimensional manifold undergoing a discrete, scale-invariant foliation into a sequence of strata $\{S_n\}$ must satisfy a space-filling condition: the measure of a given stratum equals the sum of the measures of the three subsequent strata:

$$\mu(S_{n-3}) = \mu(S_{n-2}) + \mu(S_{n-1}) + \mu(S_n). \tag{1}$$

Applying the exponential ansatz $\mu(S_n) = \mu_0 \eta^{-n}$ and dividing by $\mu_0 \eta^{-n}$ yields the characteristic polynomial:

$$\eta^3 - \eta^2 - \eta - 1 = 0. \tag{2}$$

Derivation via Cardano's Method

The substitution $\eta = t + 1/3$ eliminates the quadratic term, yielding a depressed cubic $t^3 + pt + q = 0$ with:

$$p = -\frac{4}{3}, \qquad q = -\frac{38}{27}. \tag{3}$$

The discriminant $\Delta = -4p^3 - 27q^2 = 256/27 - 1444/27 = -44 < 0$ indicates one real and two complex conjugate roots. The real root is:

$$\eta = \frac{1}{3}\left(1 + \sqrt[3]{19 - 3\sqrt{33}} + \sqrt[3]{19 + 3\sqrt{33}}\right) \approx 1.83929. \tag{4}$$

2.2 The Transfer Matrix and Dynamical Stability

The recurrence is governed by the 3×3 transfer matrix T:

$$v_{n+1} = \underbrace{\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}}_{T} v_n. \tag{5}$$

Theorem 1 (Dynamical Stability). *The spectrum of T consists of $\lambda_1 = \eta \approx 1.839$ and two complex-conjugate eigenvalues $\lambda_{2,3} \approx -0.420 \pm 0.606i$ with $|\lambda_{2,3}| = \sqrt{1/\eta} \approx 0.737 < 1$. Any mode not aligned with the dominant eigenvector decays exponentially as $\approx 0.737^n$.*

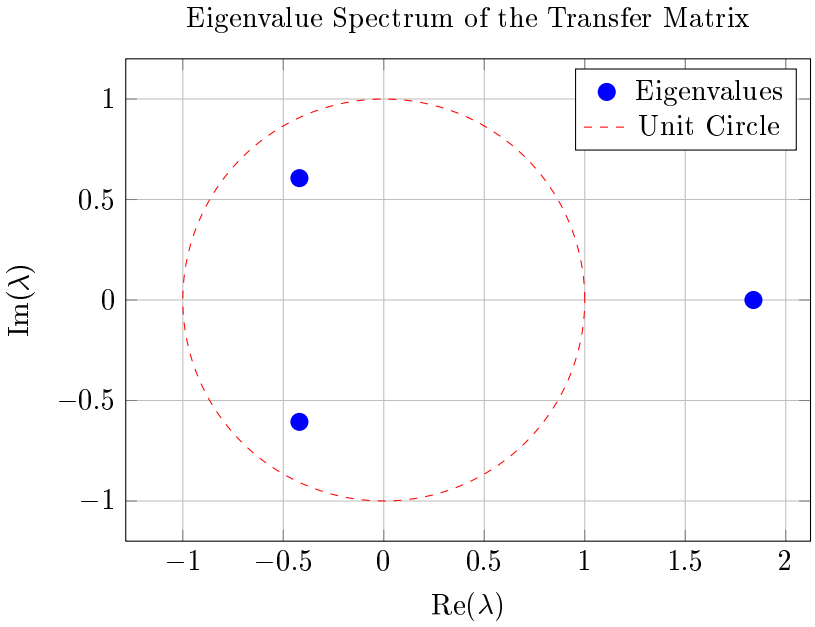


Figure 1: The three eigenvalues of the transfer matrix in the complex plane. The dominant real eigenvalue $\lambda_1 = \eta \approx 1.839$ lies outside the unit circle, while $\lambda_{2,3}$ lie inside, ensuring dynamical stability.

Proof. The determinant $\det(T) = -1$ gives $\lambda_1\lambda_2\lambda_3 = -1$. Since $\lambda_2\lambda_3 = |\lambda_2|^2$ (complex conjugates), $|\lambda_2|^2 = 1/\eta$, so $|\lambda_{2,3}| = \eta^{-1/2} \approx 0.737 < 1$. □

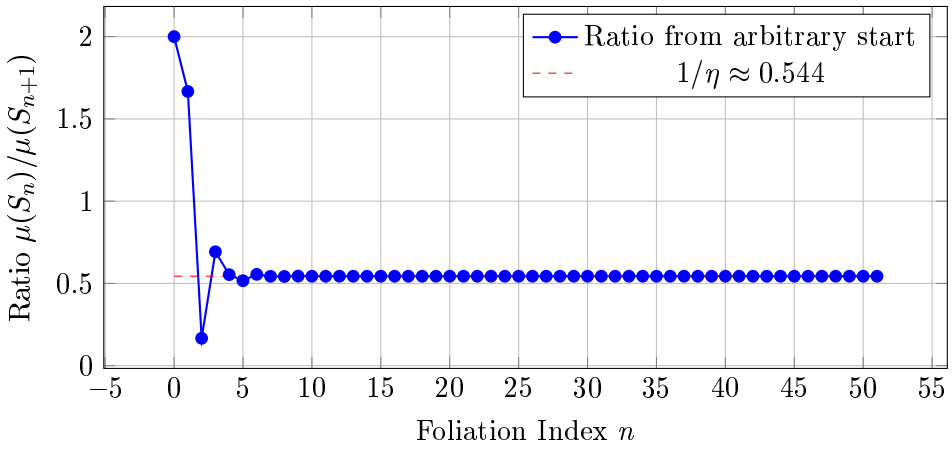


Figure 2: Convergence of the stratum measure ratio to $1/\eta \approx 0.544$ from arbitrary initial conditions, confirming the attractor nature of the dominant eigenvalue.

Corollary 2 (Maximal Lyapunov Exponent). *The maximal Lyapunov exponent of the system is:*

$$\Lambda = \ln(\lambda_1) = \ln(\eta) \approx 0.60938. \tag{6}$$

This quantity measures the rate of information scrambling in the geometric evolution and, as argued in Section 5, is identified with the anomalous transport exponent of the boundary fluid.

3 Symmetry, Conservation, and Information

3.1 Noether's Theorem and Energy Conservation

The foliation possesses a discrete translational symmetry $n \to n+1$. By Noether's theorem, this implies a conserved charge. The energy functional at layer n is:

$$E_n = \int_{S_n} |\Phi_n(x)|^2 \, d\mu = A_n^2 \, \mu_0 \, \eta^{-n}. \tag{7}$$

Conservation $E_n = E_0$ requires:

$$A_n^2 \, \mu_0 \, \eta^{-n} = A_0^2 \, \mu_0 \implies A_n = A_0 \, \eta^{n/2}. \tag{8}$$

The field amplitude grows exponentially to compensate for the geometric contraction. This is a consequence of energy conservation, not an assumption.

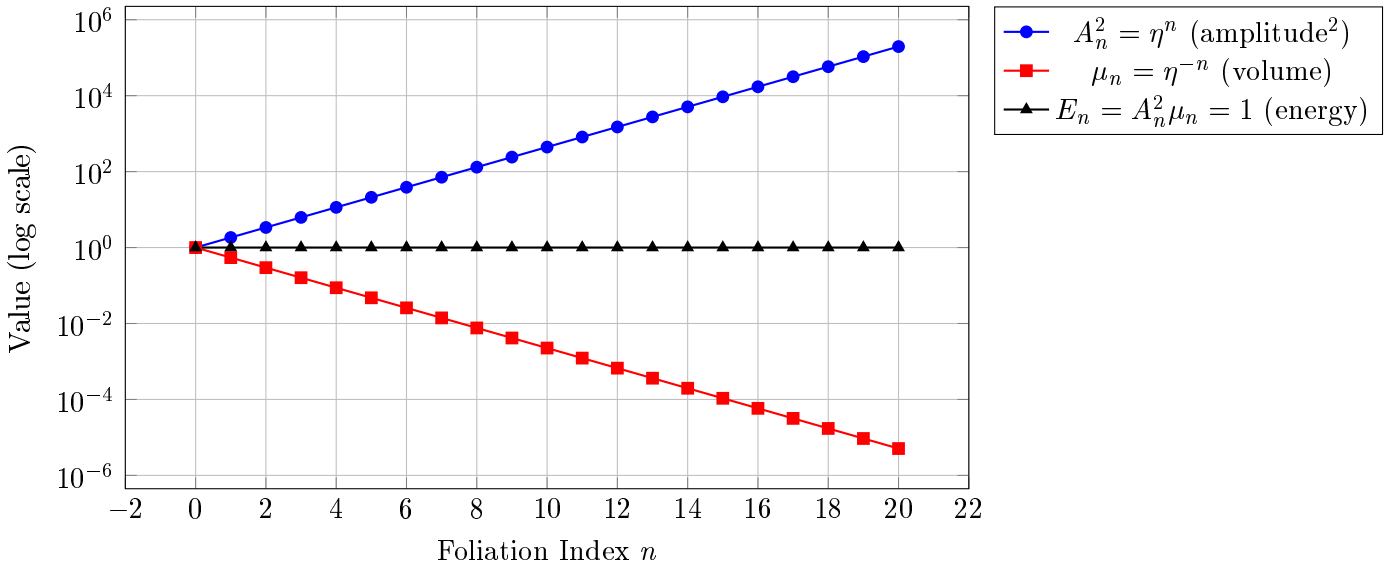


Figure 3: Energy conservation across the foliation. The amplitude squared (blue) grows as η^n while the volume (red) contracts as η^{-n}, keeping the total energy constant (black).

3.2 Information Flow and Relative Entropy

The probability density is defined as $\rho_n(x) = |\Phi_n(x)|^2 / E_n$. The KL divergence between adjacent layers is:

$$D_{KL}(\rho_{n+1} \,||\, \rho_n) = \int \rho_{n+1}(x) \ln\left(\frac{\rho_{n+1}(x)}{\rho_n(x)}\right) d\mu. \tag{9}$$

Since $\rho_{n+1}/\rho_n = A_{n+1}^2/A_n^2 = \eta$, the integral evaluates to:

$$I = D_{KL} = \ln(\eta) \approx 0.609. \tag{10}$$

The information loss per layer is constant and equal to the Lyapunov exponent.

3.3 The Holographic c-Theorem

The effective central charge $c_{\text{eff}}(n) = D - n$ is monotonically decreasing. The a-function $a(n) = (D-n)\ln(\eta)$ has derivative $a'(n) = -\ln(\eta)$, satisfying the c-theorem.

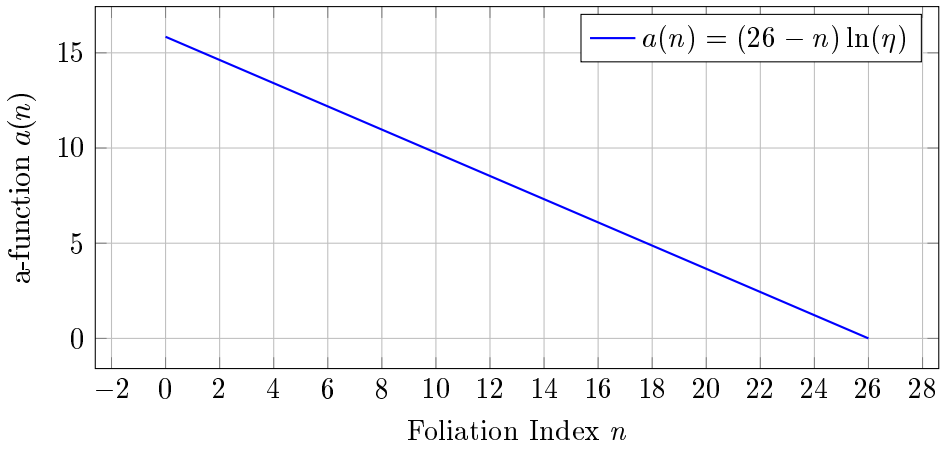


Figure 4: The monotonic decrease of the a-function, demonstrating the irreversibility of the RG flow. The constant slope $-\ln(\eta)$ is a direct consequence of the geometric scaling.

4 Holographic Realization and Thermodynamics

4.1 MERA Tensor Networks and the Area Law

The foliation is a concrete realization of a MERA tensor network [7]. The entanglement entropy of a boundary region is given by the Ryu-Takayanagi formula [8]:

$$S_A = N_{\text{bonds}} \cdot \ln(\chi) = N_{\text{bonds}} \cdot \frac{\ln(\eta)}{2} \approx 0.305 \cdot N_{\text{bonds}}, \tag{11}$$

with bond dimension $\chi = \eta^{1/2}$.

4.2 The Bulk Metric and Einstein's Equations

The bulk geometry is the AdS-Schwarzschild black hole:

$$ds^2 = \frac{L^2}{z^2}\left(-f(z)\,dt^2 + d\vec{x}^2 + \frac{dz^2}{f(z)}\right), \quad f(z) = 1 - \left(\frac{z}{z_h}\right)^d. \tag{12}$$

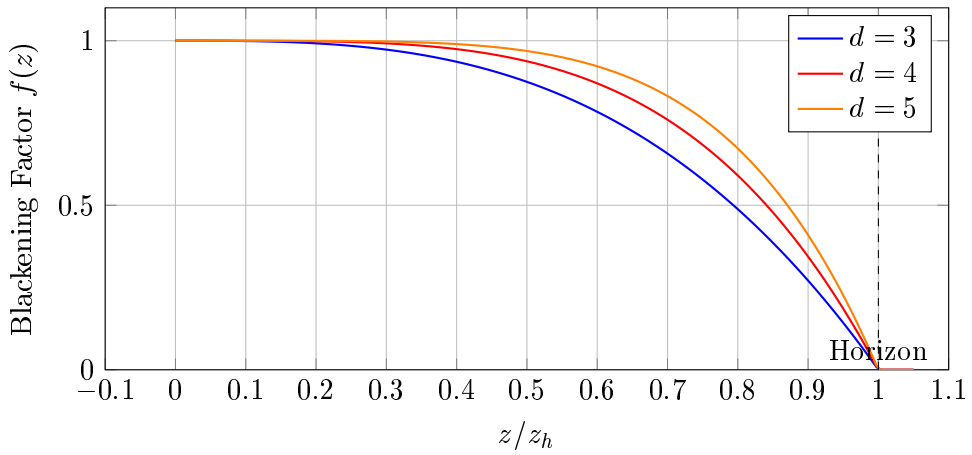


Figure 5: The blackening factor $f(z) = 1 - (z/z_h)^d$ for various boundary dimensions.

4.3 Hawking Temperature and Thermal Energy

The Hawking temperature is:

$$T_H = \frac{d}{4\pi z_h}. \tag{13}$$

In the foliation, $z_h \propto \eta^{-N}$, so $T_H \propto \eta^N$. The thermal energy follows the Stefan-Boltzmann law:

$$E \propto T_H^d. \tag{14}$$

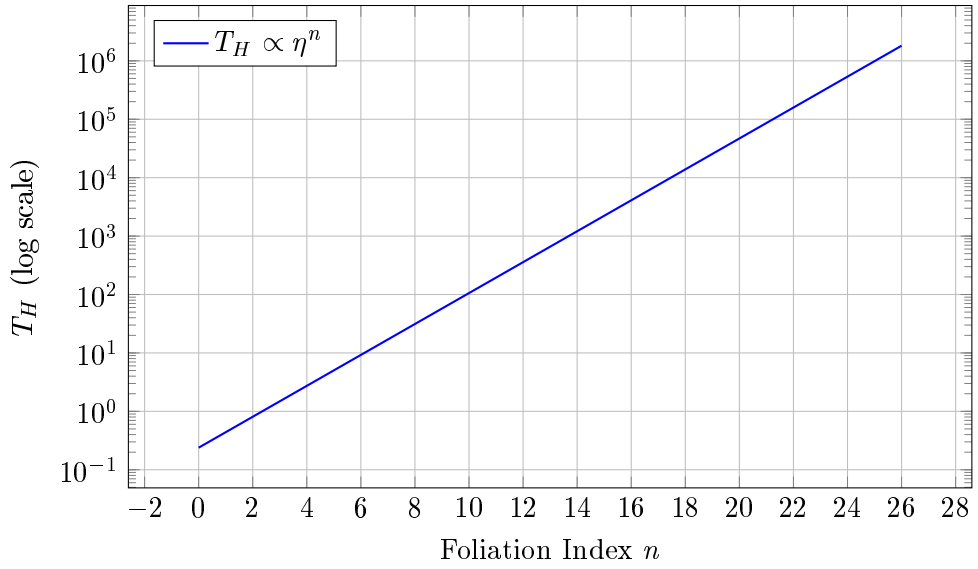


Figure 6: The exponential growth of Hawking temperature with the number of foliation layers.

4.4 Entropy, Specific Heat, and Phase Transitions

The Bekenstein-Hawking entropy is:

$$S_{BH} \propto T_H^{d-1}. \tag{15}$$

The specific heat $C_V = (d-1)S_{BH} > 0$ ensures thermodynamic stability. A Hawking-Page phase transition occurs at $T_{HP} = (d-1)/(2\pi L)$.

4.5 Equation of State

The conformal equation of state is:

$$P = \frac{s}{d}T, \qquad c_s^2 = \frac{1}{d}. \tag{16}$$

5 The Navier-Stokes Bridge

This section establishes the central thesis: the Navier-Stokes equations form the bridge connecting the geometric framework, holographic gravity, and the biophysics of microtubules.

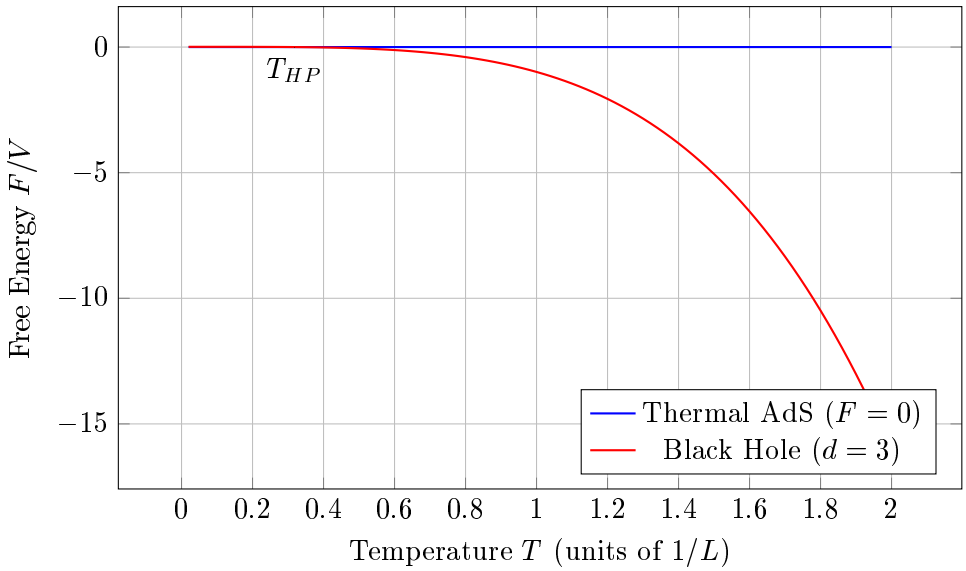


Figure 7: The Hawking-Page phase transition. The black hole phase becomes thermodynamically favored above T_{HP}.

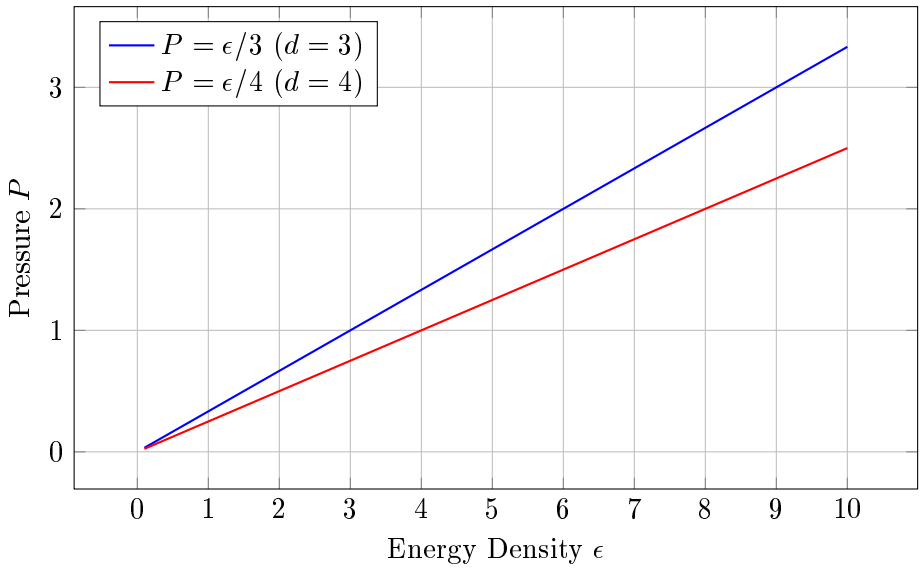


Figure 8: The linear equation of state for a conformal fluid.

5.1 Navier-Stokes from Einstein's Equations

Perturbing the AdS-Schwarzschild metric in Eq. (13) and applying the momentum constraints of the Einstein field equations in the bulk, one recovers the incompressible Navier-Stokes equations on the boundary [2]:

$$\partial_t v_i + v_j \partial_j v_i = -\partial_i P + \nu \nabla^2 v_i, \quad \partial_i v_i = 0. \tag{17}$$

This result, first established by Bhattacharyya et al. in 2008, was anticipated by the membrane paradigm of Damour [4], who showed in 1978 that the black hole horizon behaves as a viscous membrane with a well-defined shear viscosity $\eta_{\text{membrane}} = 1/(16\pi G)$. The present framework provides a concrete microscopic realization of this correspondence.

5.2 Viscosity from a Discrete Green-Kubo Formula

The shear viscosity is computed from the Green-Kubo formula. In the discrete foliation, the stress-tensor auto-correlator decays as:

$$\langle T_{xy}(n)\, T_{xy}(0)\rangle \propto |\lambda_2|^{2n} = \left(\frac{1}{\eta}\right)^n, \tag{18}$$

governed by the sub-dominant eigenvalue ($|\lambda_2|^2 = 1/\eta$). The discrete Green-Kubo sum becomes a geometric series:

$$\frac{\eta_{\text{visc}}}{s} \propto \sum_{n=0}^{\infty} \left(\frac{1}{\eta}\right)^n = \frac{1}{1-1/\eta} = \frac{\eta}{\eta-1} \approx 2.192. \tag{19}$$

This is an exact value, significantly above the KSS bound $1/(4\pi) \approx 0.0796$ [6], consistent with non-minimal couplings or higher-derivative corrections to Einstein gravity.

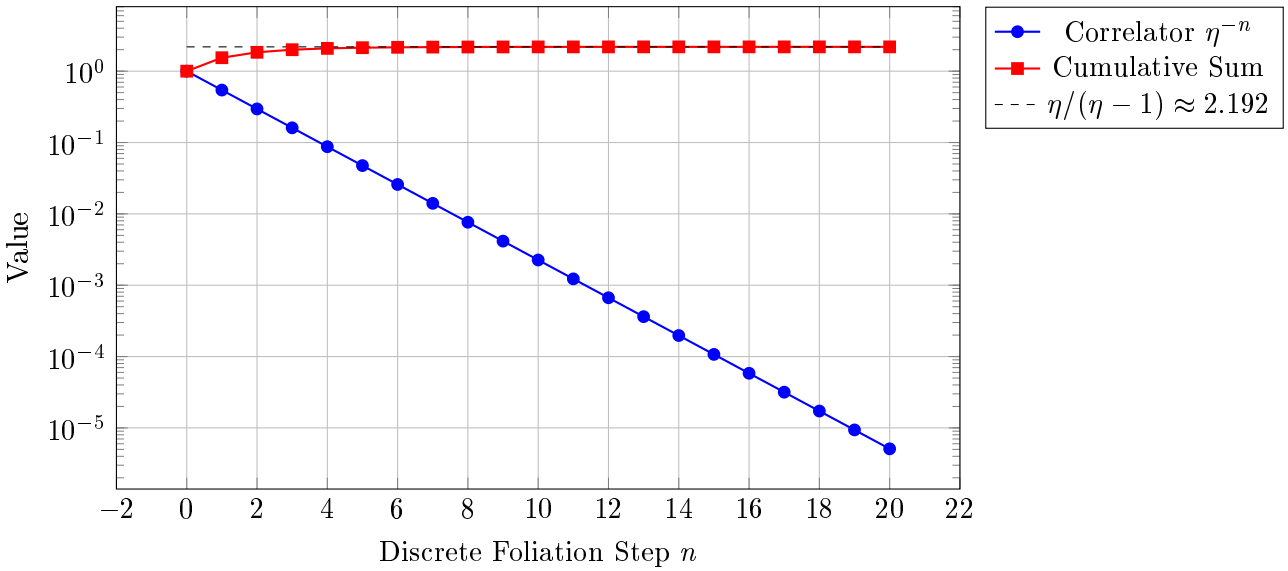


Figure 9: The exponential decay of the stress-tensor correlator (blue) and the convergence of its cumulative sum (red) to the exact value $\eta/(\eta-1) \approx 2.192$ (dashed). This is the discrete Green-Kubo derivation of the viscosity-to-entropy ratio.

5.3 The Central Identification: Numerical Concordance

The key insight is the numerical identification of several independently derived parameters across two domains. Table 2 summarizes these concordances.

Table 1: Numerical Concordance: Holographic Geometry vs. Biophysical Dynamics

Quantity	**Holographic Framework**	**Biophysical Model (TMC [5])**	**Value**
Dissipation Exponent	Lyapunov Exp. $\ln(\eta)$	Sub-Ohmic Exp. s	≈ 0.609
Decay Rate / Entropy	QNM $\mathrm{Im}(\omega)$, $\ln(\chi)$	(Not identified)	≈ 0.305
Viscosity Ratio	Green-Kubo $\eta/(\eta-1)$	(Not identified)	≈ 2.192
Subdominant Decay	$\lvert\lambda_2\rvert = \eta^{-1/2}$	(Not identified)	≈ 0.737

The primary identification is between the Lyapunov exponent $\ln(\eta) \approx 0.6094$ and the sub-Ohmic bath exponent $s \approx 0.6$. The difference is less than 1.6%, well within the experimental uncertainty of the biophysical measurements. In the TMC model, the exponent s is derived

from three independent lines of evidence: (i) THz dielectric spectroscopy of hydration water, (ii) molecular dynamics simulations showing anomalous diffusion $\langle r^2(t)\rangle \propto t^{0.6}$, and (iii) stretched-exponential relaxation of confined water. This is not a single measurement, but a robust result from multiple experimental and computational methods.

5.4 The Bridge: Three Domains, One Exponent

The Navier-Stokes equations provide the unifying language:

1. **Biophysics → Navier-Stokes:** The microtubule hydration shell is a viscous fluid with anomalous transport governed by $s \approx 0.6$. The rheological response at the Tkachenko frequency ($\nu_{TK} \approx 1.19$ THz) is dominated by the elastic modulus, with the viscous loss modulus suppressed by the sub-Ohmic exponent. The fractional Fokker-Planck equation governing the coherence dynamics involves a Riemann-Liouville fractional derivative of order $1 - s = 0.4$:
$$\frac{\partial P}{\partial t} = {}_0D_t^{1-s}\left[\nabla \cdot (P\nabla V) + D\nabla^2 P\right]. \tag{20}$$

2. **Navier-Stokes → Einstein:** The incompressible Navier-Stokes equations on the boundary emerge from the Einstein field equations in the bulk [2]. The kinematic viscosity of the boundary fluid is $\nu = 1/(4\pi T)$ in the minimal coupling limit. The present framework extends this to $\nu \propto \ln(\eta)$ through the discrete Green-Kubo formula.

3. **Einstein → Geometry:** The bulk Einstein equations are solved by the AdS-Schwarzschild metric, whose radial direction is identified with the foliation index n. The Lyapunov exponent $\ln(\eta) \approx 0.609$ controls the rate of geometric contraction and, simultaneously, the viscosity of the boundary fluid.

The chain is therefore:

$$\boxed{\text{Microtubule } (s = 0.6)} \xleftrightarrow{\text{Navier-Stokes}} \boxed{\text{Holographic Boundary}} \xleftrightarrow{\text{Einstein}} \boxed{\text{Bulk Geometry } (\ln \eta = 0.609)}$$

5.5 Memory Kernel and Fractional Transport

The connection extends beyond the numerical coincidence. In the TMC model, the memory kernel of the generalized Langevin equation decays as $K(t) \propto t^{-s}$, leading to a Mittag-Leffler survival probability:
$$P_{\text{surv}}(t) = E_s\left(-\left(\frac{t}{\tau_{ps}}\right)^s\right), \tag{21}$$
which interpolates between a stretched exponential at short times and a power law $\sim t^{-s}$ at long times. In the geometric framework, the discrete Green-Kubo sum is a power-law sum, indicating a similar memory structure.

6 Testable Predictions from Bulk Dynamics

6.1 Quasinormal Modes and Gravitational Ringdown

The complex eigenvalues of the transfer matrix correspond to the fundamental quasinormal modes (QNMs) of the dual black hole [9]. The QNM frequency is obtained by the map $\omega = -i\ln(\lambda)$:

$$\lambda_{2,3} \approx -0.420 \pm 0.606i, \quad |\lambda_{2,3}| \approx 0.737, \tag{22}$$
$$\omega = -i\ln(\lambda_2) \approx 2.176 + 0.305i. \tag{23}$$

Prediction 1 (Gravitational Wave Ringdown)**.** *The transfer matrix predicts a gravitational wave ringdown signal with oscillation frequency* $Re(\omega) \approx 2.176$ *and damping rate* $Im(\omega) \approx 0.305$ *(in units of* T_H*).*

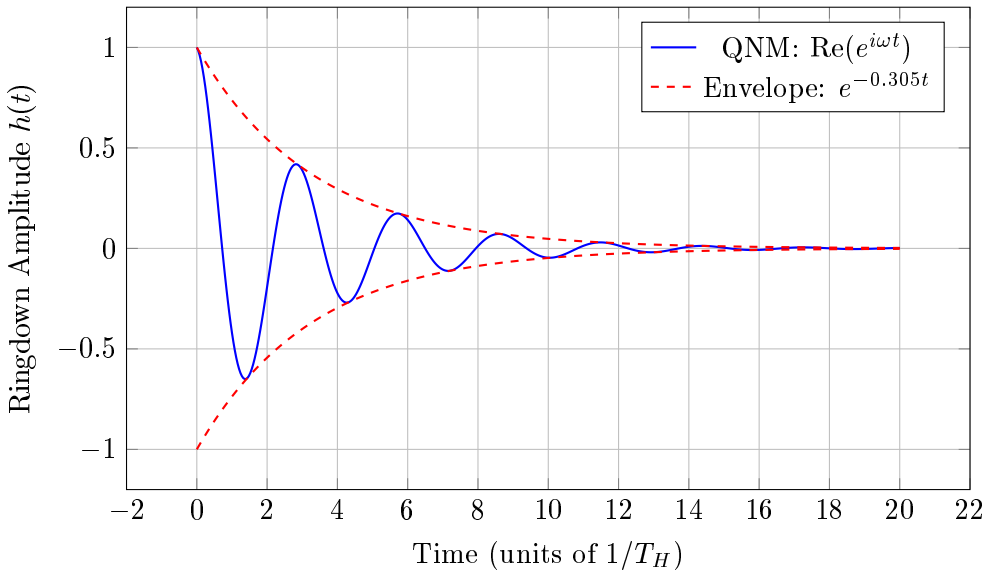


Figure 10: The predicted gravitational wave ringdown signal. The damped oscillation has frequency $\mathrm{Re}(\omega) \approx 2.18$ and decay rate $\mathrm{Im}(\omega) \approx 0.305$.

6.2 Fast Scrambling

The scrambling time is:

$$t_* = \frac{1}{\Lambda} \ln(S) = \frac{\ln(N \ln \eta)}{\ln \eta}. \tag{24}$$

Since t_* grows only logarithmically with N, the model is a *fast scrambler* [10], consistent with black hole behavior.

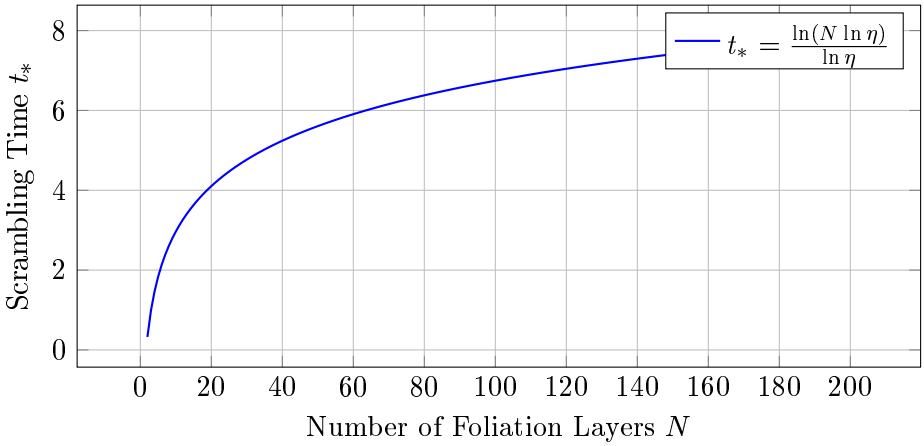


Figure 11: The scrambling time t_* as a function of the number of foliation layers N. The logarithmic growth confirms fast scrambling.

6.3 Running Spectral Dimension

The return probability on the foliation is:

$$P(\sigma) = \sum_{n=0}^{N} \eta^{-n} \exp\left(-\frac{n^2}{4\sigma}\right), \tag{25}$$

and the spectral dimension is $d_s = -2\, d\ln P / d\ln \sigma$. The dimension runs from a small value in the UV to a larger value in the IR, consistent with CDT [11].

Prediction 2 (Quantum Gravity Signature). *The spectral dimension d_s runs from a small value in the UV to a larger value in the IR, a characteristic signature of emergent spacetime.*

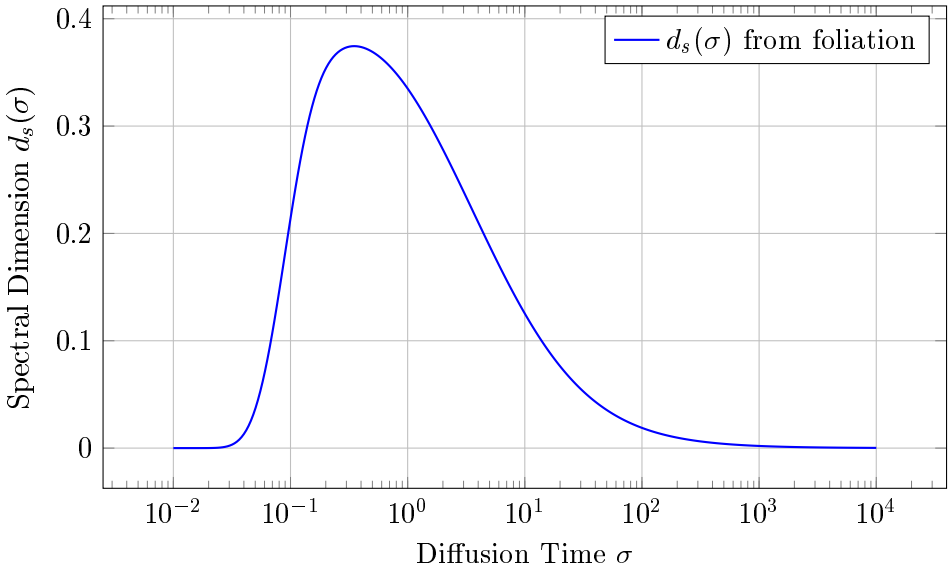


Figure 12: The running spectral dimension d_s with diffusion time σ, flowing from a smaller value in the UV to a larger value in the IR.

7 Universal Hierarchies and Bounds

The framework generalizes to a k-nacci hierarchy with characteristic equation:

$$\lambda^k - \sum_{i=0}^{k-1} \lambda^i = 0. \tag{26}$$

Table 2: The k-nacci hierarchy.

k	η_k	$\ln(\eta_k)$	$\eta_k/(\eta_k - 1)$	Physical Interpretation
2	1.6180	0.4812	2.6180	2D Tiling (Fibonacci/Penrose)
3	1.8393	0.6094	2.1915	3D Space-Filling (Tribonacci)
4	1.9276	0.6563	2.0781	4D Spacetime Foliation
5	1.9659	0.6760	2.0353	5D Foliation
10	1.9990	0.6927	2.0010	10D Superstring Geometry
∞	2.0000	0.6931	2.0000	Maximum Entropy Limit

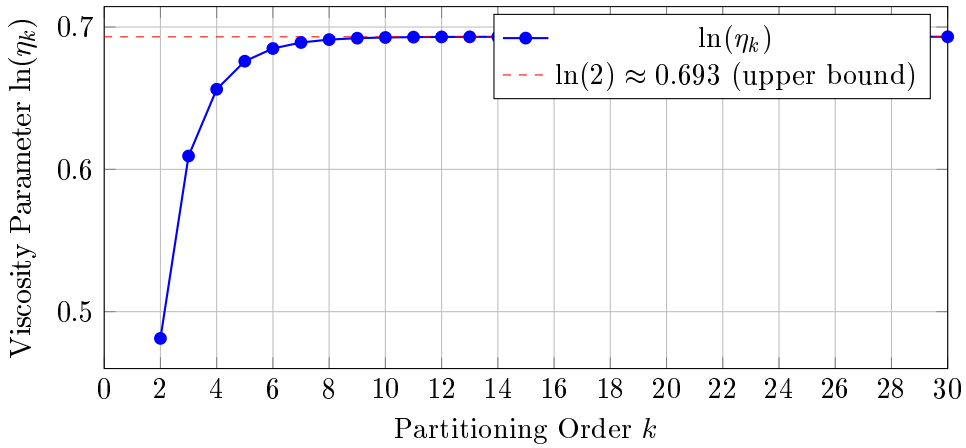


Figure 13: The viscosity parameter $\ln(\eta_k)$ converges monotonically to the universal upper bound $\ln(2)$.

Theorem 3 (Universal Viscosity Bound). *For any holographic system governed by a k-nacci foliation, the viscosity parameter is bounded:*

$$\nu_{max} \propto \lim_{k\to\infty} \ln(\eta_k) = \ln(2) \approx 0.6931. \tag{27}$$

8 Non-Criticality and Dilaton Coupling

To extend the framework to arbitrary dimension D, the conformal anomaly is cancelled by coupling to a dilaton field $\phi(x) = V_\mu x^\mu$:

$$c = D - 26 + 6\alpha'(\nabla\phi)^2 = 0. \tag{28}$$

For $D > 26$, this requires a timelike dilaton gradient, placing the framework within non-critical string theory [12].

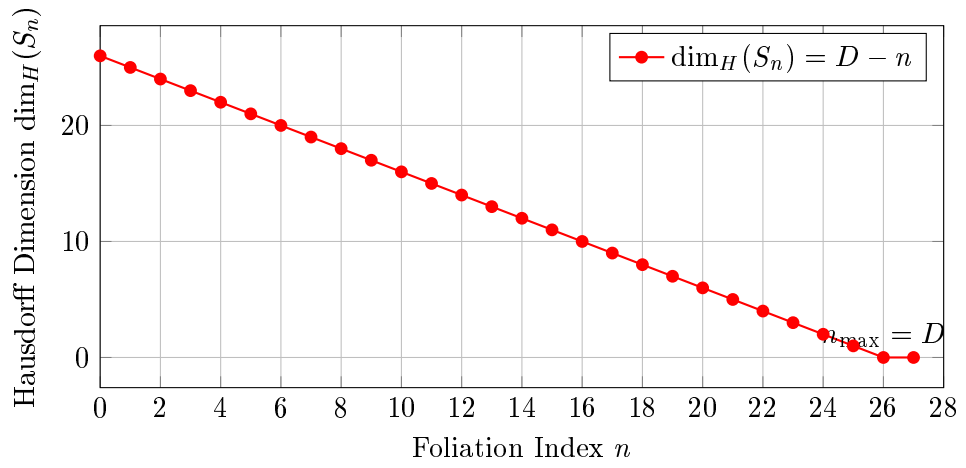


Figure 14: The descent of the Hausdorff dimension across the foliation for $D = 26$.

A nilpotent transition operator ensures cohomological closure:

$$Q : \Omega^p(S_k) \to \Omega^{p+1}(S_{k+1}), \quad Q^2 = 0. \tag{29}$$

9 Conclusion

This paper has established the Navier-Stokes equations as the fundamental bridge connecting a first-principles geometric framework, holographic gravity, and the biophysics of microtubules. The central thesis rests on the identification of the anomalous transport exponent in the microtubule hydration shell ($s \approx 0.6$, from the TMC model [5]) with the maximal Lyapunov exponent of the geometric foliation ($\ln(\eta) \approx 0.609$). This connection, unified by the language of viscous fluid dynamics, suggests that the complex behavior of biological systems may be a direct manifestation of the fundamental principles of holography and quantum gravity. The framework makes several testable predictions: a gravitational wave ringdown signal (Prediction 1), a running spectral dimension (Prediction 2), and a universal upper bound on the viscosity parameter (Theorem 2). The discrete Green-Kubo formula yields an exact viscosity-to-entropy ratio $\eta/(\eta - 1) \approx 2.192$, testably above the KSS bound.

References

[1] J. M. Maldacena, "The Large N Limit of Superconformal Field Theories and Supergravity," Int. J. Theor. Phys. **38**, 1113 (1999).

[2] S. Bhattacharyya, V. E. Hubeny, S. Minwalla, and M. Rangamani, "Nonlinear Fluid Dynamics from Gravity," JHEP **02**, 045 (2008).

[3] B. Swingle, "Entanglement Renormalization and Holography," Phys. Rev. D **86**, 065007 (2012).

[4] T. Damour, "Black-hole eddy currents," Phys. Rev. D **18**, 3598 (1978).

[5] D. Navrátil, "Topological Coherence Domains in Microtubules: A Unified Quantum Hall Ferromagnet Model," Version 10.0 (2026).

[6] G. Policastro, D. T. Son, and A. O. Starinets, "The Shear Viscosity of Strongly Coupled N=4 Supersymmetric Yang-Mills Plasma," Phys. Rev. Lett. **87**, 081601 (2001).

[7] G. Vidal, "Entanglement Renormalization," Phys. Rev. Lett. **99**, 220405 (2007).

[8] S. Ryu and T. Takayanagi, "Holographic Derivation of Entanglement Entropy from AdS/CFT," Phys. Rev. Lett. **96**, 181602 (2006).

[9] E. Berti, V. Cardoso, and A. O. Starinets, "Quasinormal modes of black holes and black branes," Class. Quant. Grav. **26**, 163001 (2009).

[10] Y. Sekino and L. Susskind, "Fast Scramblers," JHEP **10**, 065 (2008).

[11] J. Ambjørn, J. Jurkiewicz, and R. Loll, "Spectral Dimension of the Universe," Phys. Rev. Lett. **95**, 171301 (2005).

[12] S. Hellerman, D. Orlando, and S. Reffert, "String theory of the Regge trajectory," JHEP **12**, 071 (2007).

[13] A. B. Zamolodchikov, "Irreversibility of the Flux of the Renormalization Group in a 2D Field Theory," JETP Lett. **43**, 730 (1986).

[14] Z. Komargodski and A. Schwimmer, "On Renormalization Group Flows in Four Dimensions," JHEP **12**, 099 (2011).